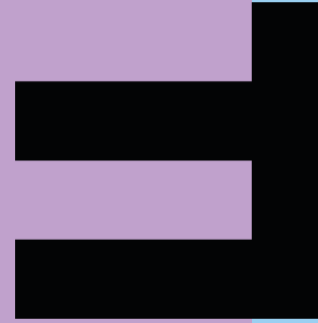


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Memories

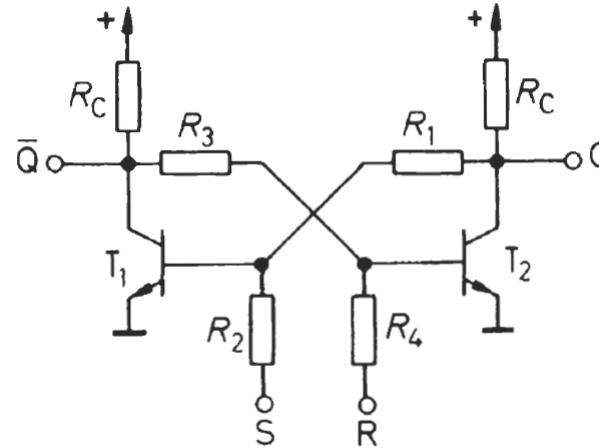
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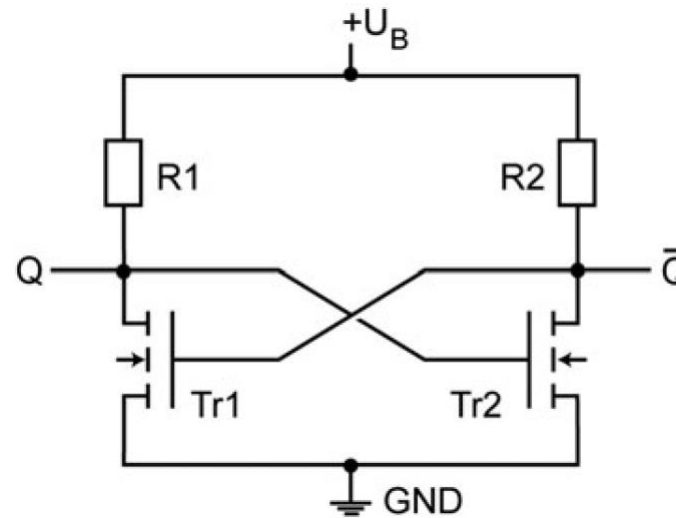
Volatile Memories

- Another type of flip-flop:



[1], S. 600

- and another:

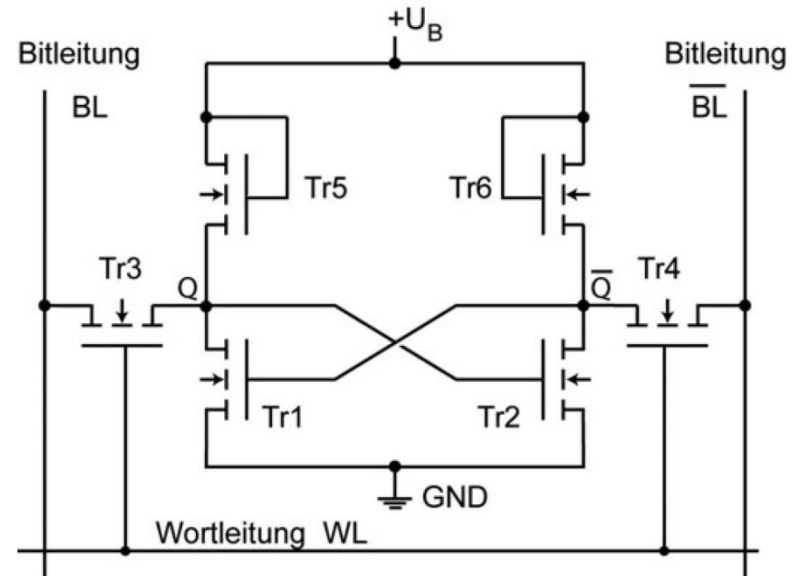


[2], S. 39



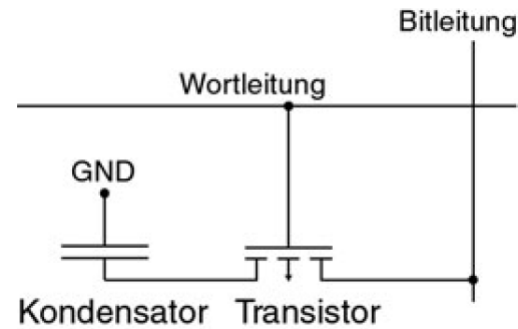
SRAM / DRAM

- A CMOS SRAM cell:



[2], S. 40

- A CMOS DRAM cell:



[2], S. 41



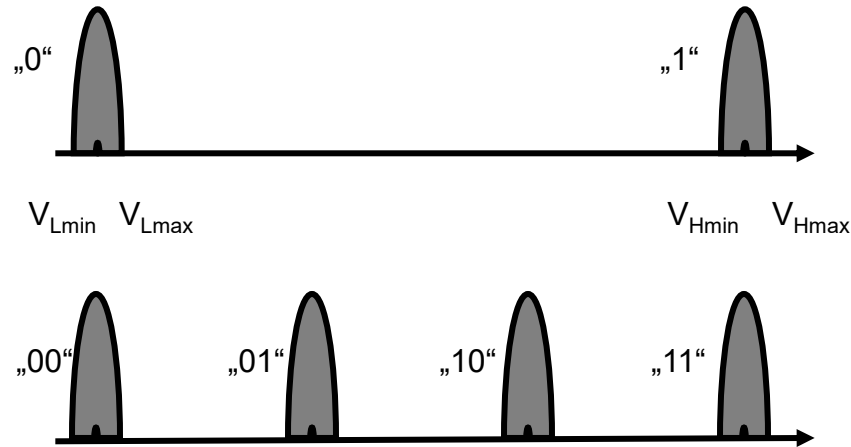
Non-volatile Memories

- ROM
- PROM
- EPROM
- E²PROM = EEPROM
- Flash



Flash Memories

- Cell voltage of single-level / multi-level cell flash memories



- Erasing flash and E²PROM memories
- Alternative: NVRAM?
- Alternative: FRAM / MRAM?



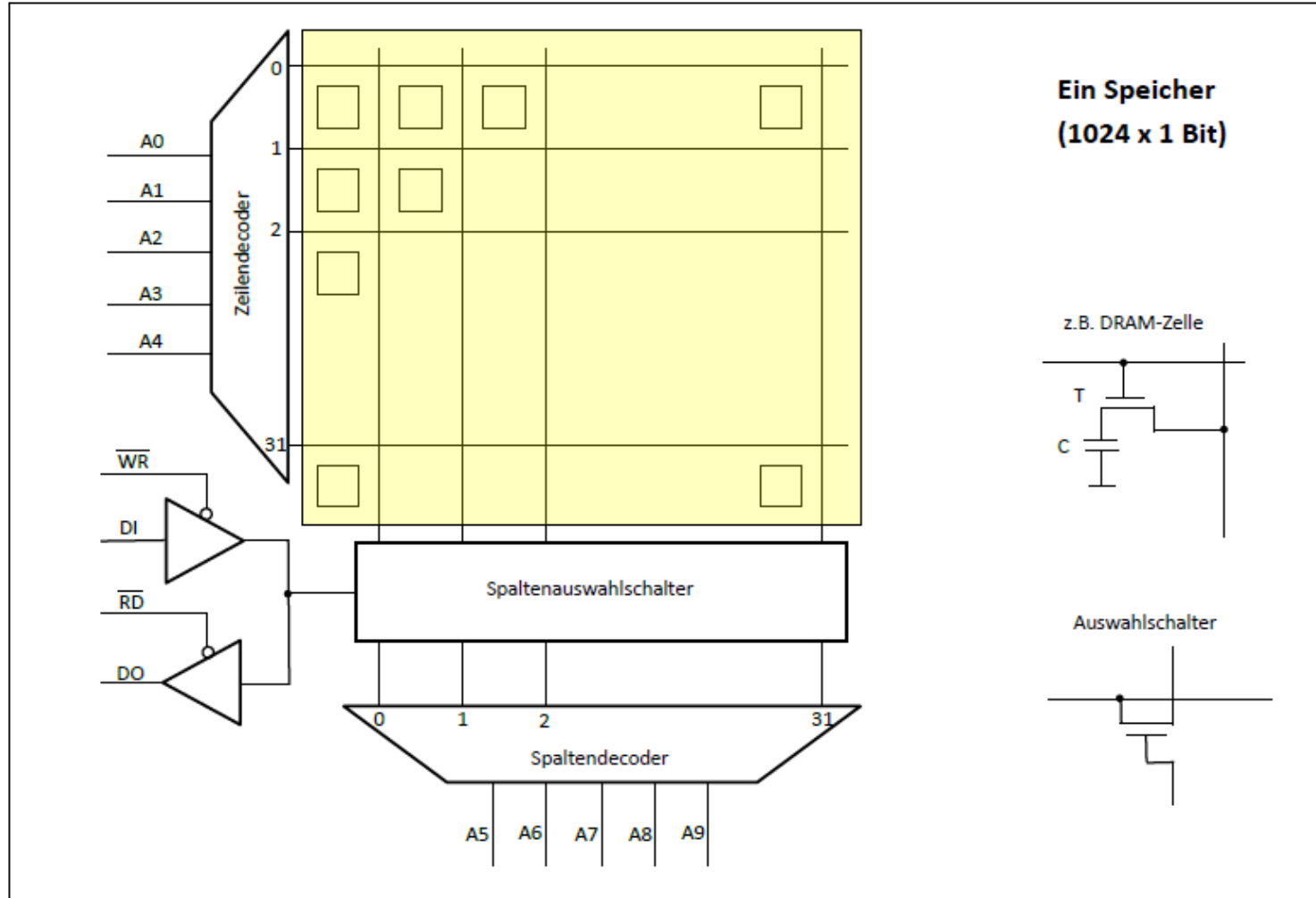
Non-volatile Memories II

- A comparison of current non-volatile memories

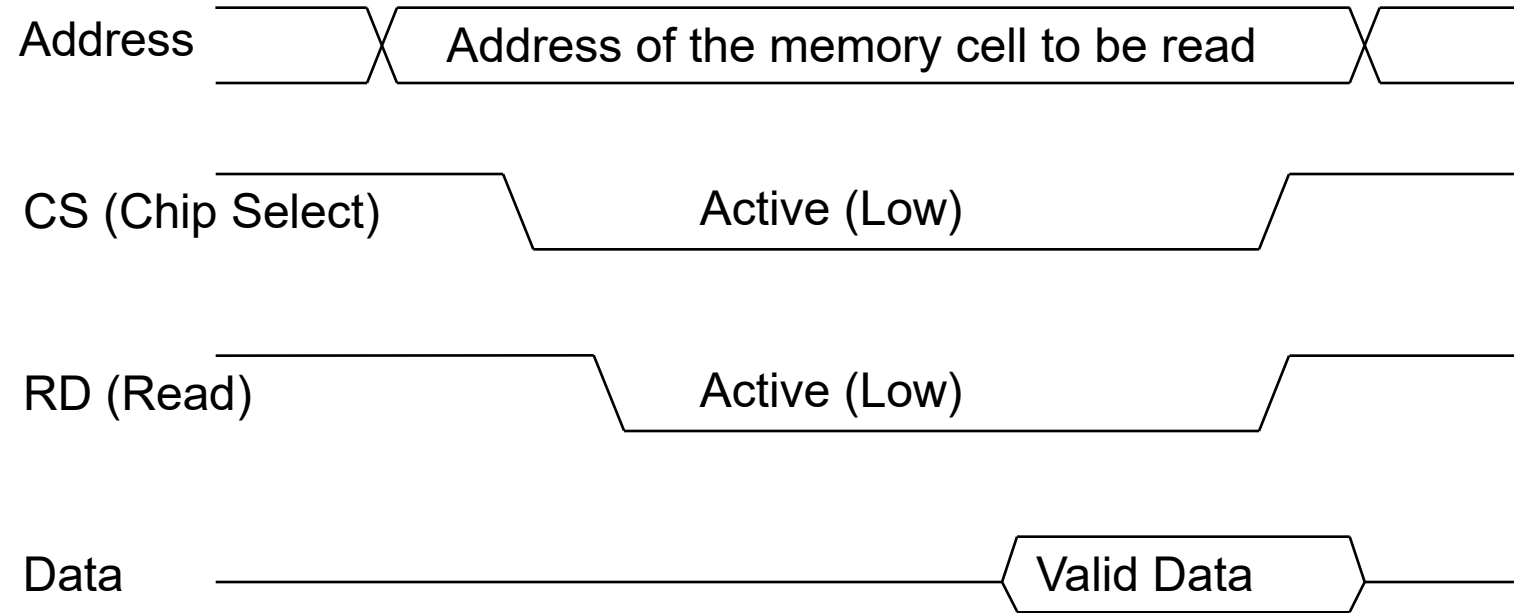
	EEPROM	Flash	SRAM + Battery	FRAM	MRAM
Memory size	1Mb	4Gb	64Mb	4Mb	4Mb
Write speed	> 1ms	> 10 μ s	~ 50ns	~ 150ns	~ 50ns
Cost	o	+	o / -	-	-
Size	+	+	+ / -	+	+
Write cycles	> 100k	> 10k	∞	> 10 ¹² / 40 Jahre	∞ / 20 Jahre



Functional Principle of a Memory



A Simple Memory Access



- Write access is similar: WR instead of RD and the data in the other direction



References

- [1] Tietze, Ulrich; Schenk, Christoph (1999): Halbleiterschaltungstechnik. 11. Auflage. Berlin: Springer
- [2] Wüst, Klaus (2011): Mikroprozessortechnik, 4. Auflage, Wiesbaden: Vieweg+Teubner



Memories

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